The documentation and process conversion measures necessary to comply with this document shall be completed by 8 April 2025.

INCH-POUND

MIL-PRF-19500/502H
w/AMENDMENT 3
8 December 2024
SUPERSEDING
MIL-PRF-19500/502H
w/AMENDMENT 2
24 May 2022

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, DARLINGTON, NPN, SILICON, POWER,
ENCAPSULATED (THROUGH-HOLE MOUNT PACKAGE)
* TYPES 2N6058 AND 2N6059, QUALITY LEVELS JAN, JANTX, JANTXV, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for NPN silicon power Darlington transistors. Three levels of product assurance are (JAN, JANTX, and JANTXV) provided for each encapsulated device. Two levels of product assurance are provided for unencapsulated die (JANHC and JANKC).

* 1.2 Package outlines. The device package outlines are as follows: TO-204AA (similar to TO-3) in accordance with figure 1 for all encapsulated device types. The dimensions and topography for JANHC and JANKC unencapsulated die are in accordance with figures 2.

1.3 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Types | $P_T$ | | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $I_B$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|
| | $T_C$ = +25°C (1) | $T_C$ = +100°C | | | | | | |
| | W | W | V dc | V dc | V dc | A dc | A dc | °C |
| 2N6058 | 150 | 75 | 80 | 80 | 5 | 12 | 0.2 | -55 to +175 |
| 2N6059 | 150 | 75 | 100 | 100 | 5 | 12 | 0.2 | -55 to +175 |

(1) Derate linearly at 1.00 W/°C above $T_C$ > +25°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, $T_C$ = +25°C.

| Limit | $h_{FE2}$ (1) $V_{CE}$ = 3 V dc $I_C$ = 6 A dc | $h_{FE3}$ (1) $V_{CE}$ = 3 V dc $I_C$ = 12 A dc | $h_{fe}$ $V_{CE}$ = 3 V dc $I_C$ = 5 A dc f = 1 kHz | $\|h_{fe}\|$ $V_{CE}$ = 3 V dc $I_C$ = 5 A dc f = 1 MHz | $C_{obo}$ 100 kHz ≤ f ≤ 1 MHz $V_{CB}$ = 10 V dc $I_E$ = 0 | $R_{\theta JC}$ |
|---|---|---|---|---|---|---|
| | | | | | pF | °C/W |
| Minimum | 1,000 | 150 | 1,000 | 10 | | |
| Maximum | 18,000 | | | 250 | 300 | 1.0 |

| Limit | $V_{BE(sat)}$ (1) $I_C$ = 12 A dc $I_B$ = 120 mA dc | $V_{CE(sat)1}$ (1) $I_C$ = 12 A dc $I_B$ = 120 mA dc | $V_{CE(sat)2}$ (1) $I_C$ = 6 A dc $I_B$ = 24 mA dc | Pulse response (2) | |
|---|---|---|---|---|---|
| | | | | $t_{on}$ | $t_{off}$ |
| | V dc | V dc | V dc | µs | µs |
| Minimum | | | | | |
| Maximum | 4.0 | 3.0 | 2.0 | 2 | 10 |

(1) Pulsed (see 4.5.1).
* (2) See figure 3 for pulse response circuits.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

* 1.5.1.1 Quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", and "JANTXV".

* 1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices that are applicable for this specification sheet are "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

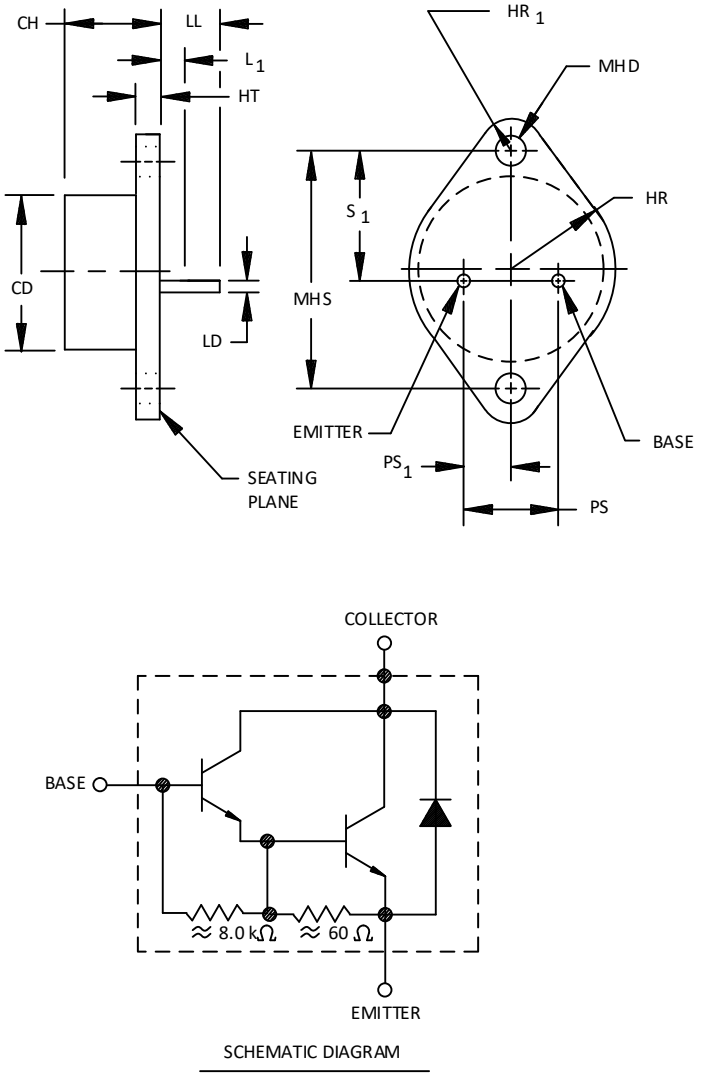
1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6058" and "6059".

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

* 1.5.4 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A"(see 6.5.2 and figure 2,).



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(Correction in Schematic)

* FIGURE 1. Physical dimensions and schematic circuit (TO-204AA, similar to TO-3).



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| Symbol | Inches | Inches | mm | mm | Notes |
|---|---|---|---|---|---|
| | min | max | min | max | |
| CD | | .875 | | 22.23 | |
| CH | .250 | .328 | 6.35 | 8.33 | |
| HR | .495 | .525 | 12.57 | 13.34 | |
| $HR_1$ | .131 | .188 | 3.33 | 4.78 | 3 |
| HT | .060 | .135 | 1.52 | 3.43 | |
| LD | .038 | .043 | 0.97 | 1.09 | 4, 5 |
| LL | .312 | .500 | 7.92 | 12.7 | 4 |
| $L_1$ | | .050 | | 1.27 | 4, 5 |
| MHD | .151 | .161 | 3.84 | 4.09 | 6 |
| MHS | 1.177 | 1.197 | 29.90 | 30.40 | |
| PS | .420 | .440 | 10.67 | 11.18 | 7, 8 |
| $PS_1$ | .205 | .225 | 5.21 | 5.72 | 7, 4, 8 |
| $S_1$ | .655 | .675 | 16.64 | 17.15 | 7 |

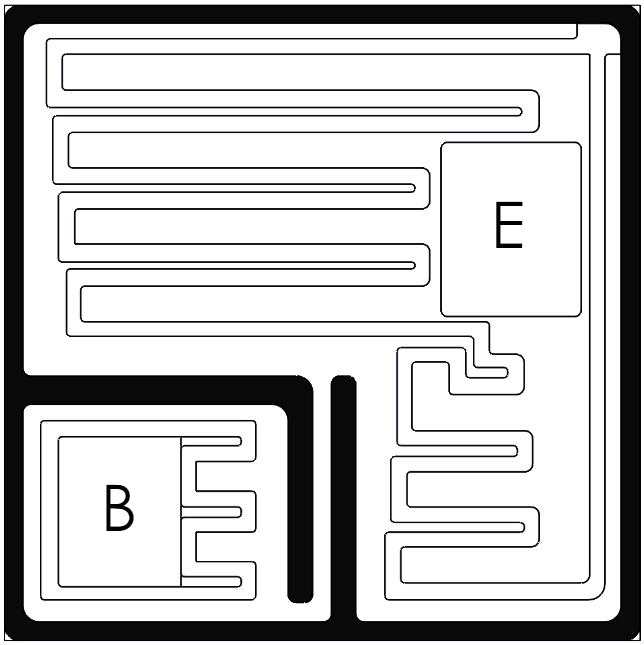
NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Body contour is optional within zone defined by dimension CD.
3. At both ends.
4. Both terminals.
5. Dimension LD applies between dimension $L_1$ and LL. Lead diameter shall not exceed twice dimension LD within dimension $L_1$. Diameter is uncontrolled in dimension $L_1$.
6. Two holes.
7. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below the seating plane. When gauge is not used, measurement shall be made at seating plane.
8. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
9. The collector shall be electrically connected to the case.

* FIGURE 1. Physical dimensions and schematic circuit (TO-204AA, similar to TO-3) - Continued.



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NOTES
1. Chip size ..... 0.198 x 0.198 inch ±.002 inch (5.029 x 5.029 mm ±0.05 mm).
2. Chip thickness..... 0.014 inch ±.002 inch (0.36 mm ±.05 mm).
3. Top metal..... Aluminum 51,000 Å minimum, 60,000Å nominal.
4. Back metal ..... Ti/Ni/Ag 8,050 Å minimum, 11,500Å nominal.
5. Backside .....Collector.
6. Bonding pad.....B = 0.039 x 0.047 inch (0.991 x 1.194 mm).
E = 0.043 x 0.054 inch (1.092 x 1.372 mm).

* FIGURE 2. Physical dimensions, JANHC and JANKC die (A Version).



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

* 3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-204AA, similar to TO-3) and figure 2 herein.

3.4.1 Lead finish. The lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Polarity. The polarity of the device types shall be as shown on figure 1. The collector shall be electrically connected to the case.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.



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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

* 4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurements for quality levels JANTX and JANTXV only |
|---|---|
| * 3c (1) | Thermal impedance, see 4.3.2 |
| 9 | $I_{CEX1}$. |
| * 11 | $I_{CEX1}$, $h_{FE2}$; $\Delta I_{CEX1}$ = 100 percent of initial value or 2 µA dc, whichever is greater. |
| 12 | See 4.3.3. |
| 13 | See table I, subgroup 2 herein.<br>$\Delta I_{CEX1}$ = 100 percent of initial value or 2 µA dc, whichever is greater.<br>$\Delta h_{FE2}$ = ±40 percent of initial value. |

(1) This test shall be performed anytime after temperature cycling, screen 3a, and does not need to be repeated in screening requirements.

* 4.3.1 Screening of unencapsulated die (JANHC and JANKC). Screening of JANHC and JANKC unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.



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* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with test method 3131 of MIL-STD-750 using the guidelines in that test method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, $t_{MD}$ (and $V_H$ where appropriate). The thermal impedance limit used in screen 3c and table I, subgroup 2 herein shall be set statistically by the supplier. See table II, subgroup 4 herein.

* 4.3.3 Power burn-in conditions. The power burn-in conditions shall be as follows: $T_J$ = +162.5°C ± 12.5°C; $V_{CE}$ ≥ 10 V dc, $T_A$ ≤ +100°C. NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Delta requirements shall be in accordance with the applicable step of 4.7 herein.

4.4.2.1 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

| | Subgroup | Method | Conditions |
|---|---|---|---|
| | B3 | 1037 | $V_{CB}$ ≥ 10 V dc; $\Delta T_J$ = between cycles ≥ +100°C. $t_{on}$ = $t_{off}$ = 3 minutes. No heat sink or forced-air cooling on the devices shall be permitted. |
| * | B5 | 3131 | See 4.3.2, $R_{\theta JC}$ = 1°C/W (maximum). |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Delta requirements shall be in accordance with the applicable step of 4.7 herein.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Test condition A, weight = 10 lbs (4.54 Kg), t = 15s. |
| C6 | 1037 | $V_{CB}$ ≥ 10 V dc; $\Delta T_J$ between cycles ≥ +100°C; $t_{on}$ = $t_{off}$ = 3 minutes. No heat sink or forced-air cooling on device shall be permitted. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Delta requirements shall be in accordance with the applicable step of 4.7 herein.



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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse response measurements. The conditions for pulse response measurement shall be as specified in section 4 of MIL-STD-750.

* 4.6 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500.

* 4.7 Delta measurements. The requirements for delta measurements for groups B, C, and E inspection shall be as specified below. (1) (2) (3)

| Step | Inspection | MIL-STD-750 | | Symbol | Limit |
|---|---|---|---|---|---|
| | | Method | Conditions | | |
| 1 | Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 6.0 A dc; pulsed (see 4.5.1). | $\Delta h_{FE2}$ | ±40 percent change from initial reading. |

(1) The delta measurement for group B, quality levels (JAN, JANTX and JANTXV) shall be as follows: In addition to the measurements specified for subgroups 3 and 6 of table E-VIB of MIL-PRF-19500, the measurements of step 1 shall also be taken.
(2) The delta measurement for group C shall be as follows: In addition to the measurements specified for subgroup 6 of table E-VII of MIL-PRF-19500, the measurements of step 1 shall also be taken.
(3) The delta measurement for group E shall be as follows: In addition to the measurements specified for subgroups 1 and 2 of table E-IX of MIL-PRF-19500, the measurements of step 1 shall also be taken.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| * Thermal impedance 2/ | 3131 | See 4.3.2 | $Z_{\theta JC}$ | | | °C/W |
| Collector to emitter breakdown voltage | 3011 | Bias condition D, $I_C$ = 100 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N6058 | | | | 80 | | V dc |
| 2N6059 | | | | 100 | | V dc |
| * Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX1}$ | | | |
| 2N6058 | | $V_{CE}$ = 80 V dc | | | 10 | µA dc |
| 2N6059 | | $V_{CE}$ = 100 V dc | | | 10 | µA dc |
| Collector to emitter cutoff current | 3041 | Bias condition D | $I_{CEO}$ | | | |
| 2N6058 | | $V_{CE}$ = 40 V dc | | | 1.0 | mA dc |
| 2N6059 | | $V_{CE}$ = 50 V dc | | | 1.0 | mA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 5 V dc | $I_{EBO}$ | | 2.0 | mA dc |
| Base to emitter voltage (non-saturated) | 3066 | Test condition B, $V_{CE}$ = 3 V dc; $I_C$ = 6 A dc | $V_{BE}$ | | 2.8 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A, $I_C$ = 12 A dc; $I_B$ = 120 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)}$ | | 4.0 | V dc |
| Collector to emitter voltage (saturation voltage) | 3071 | $I_C$ = 12 A dc; $I_B$ = 120 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 3.0 | V dc |
| Collector to emitter voltage (saturation voltage) | 3071 | $I_C$ = 6 A dc; $I_B$ = 24 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 2.0 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 1 A dc; pulsed (see 4.5.1) | $h_{FE1}$ | 1,000 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 6 A dc; pulsed (see 4.5.1) | $h_{FE2}$ | 1,000 | 18,000 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 12 A dc; pulsed (see 4.5.1) | $h_{FE3}$ | 150 | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX2}$ | | | |
| 2N6058 | | $V_{CE}$ = 80 V dc | | | 5.0 | mA dc |
| 2N6059 | | $V_{CE}$ = 100 V dc | | | 5.0 | mA dc |
| Collector to emitter voltage (saturation voltage) | 3071 | $I_C$ = 6 A dc; $I_B$ = 24 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)3}$ | | 2.0 | V dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 6 A dc; pulsed (see 4.5.1) | $h_{FE4}$ | 300 | | |
| Subgroup 4 | | | | | | |
| Small-signal, short-circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 3 V dc; $I_C$ = 5 A dc; f = 1 kHz | $h_{fe}$ | 1,000 | | |
| Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 3 V dc; $I_C$ = 5 A dc; f = 1.0 MHz | $\|h_{fe}\|$ | 10 | 250 | |
| Open-circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 300 | pF |
| Pulse response | 3251 | | | | | |
| * Turn-on time | | (See figure 3), $V_{CC}$ = 30 V dc; $I_C$ = 5 A dc; $I_B$ = 20 mA dc | $t_{on}$ | | 2.0 | µs |
| Turn-off time | | (See figure 2); $V_{CC}$ = 30 V dc; $I_C$ = 5 A dc; $I_{B1}$ = $I_{B2}$ = 20 mA dc | $t_{off}$ | | 10 | µs |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| * Safe operating area (continuous dc) | 3051 | $T_C$ = +25°C +10°C, -0°; t ≥ 1 s; 1 cycle; (see figure 4) | | | | |
| Test 1 (Both device types) | | $V_{CE}$ = 12.5 V dc; $I_C$ = 12 A dc | | | | |
| Test 2 (Both device types) | | $V_{CE}$ = 30 V dc; $I_C$ = 5 A dc | | | | |
| Test 3 | | | | | | |
| (2N6058) | | $V_{CE}$ = 70 V dc; $I_C$ = 200 mA dc | | | | |
| (2N6059) | | $V_{CE}$ = 90 V dc; $I_C$ = 155 mA dc | | | | |
| Safe operating area (switching) | 3053 | Load condition B; (clamped inductive load); $T_A$ = +25°C; $t_r + t_f$ ≤ 1.0 µs; duty cycle ≤ 2 percent; $t_p$ = 1 ms; (vary to obtain $I_C$); $R_s$ = 0.10 Ω; $R_{BB1}$ = 80 Ω; $V_{BB1}$ = 16 V dc; $R_{BB2}$ = 100 Ω; $V_{BB2}$ = 1.5 V dc; $I_C$ = 12 A dc; $V_{CC}$ = 20 V dc; $R_L$ ≤ 2 Ω; L = 10 mH (Stancor C-2688 or equivalent); (see figure 5 for reference only). | | | | |
| * 2N6058 | | Clamp voltage = 80 +0, -5 V dc | | | | |
| 2N6059 | | Clamp voltage = 100 +0, -5 V dc | | | | |
| | | Device fails if clamp voltage not reached. | | | | |
| End-point electrical measurements | | See subgroup 2 of table I herein. | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



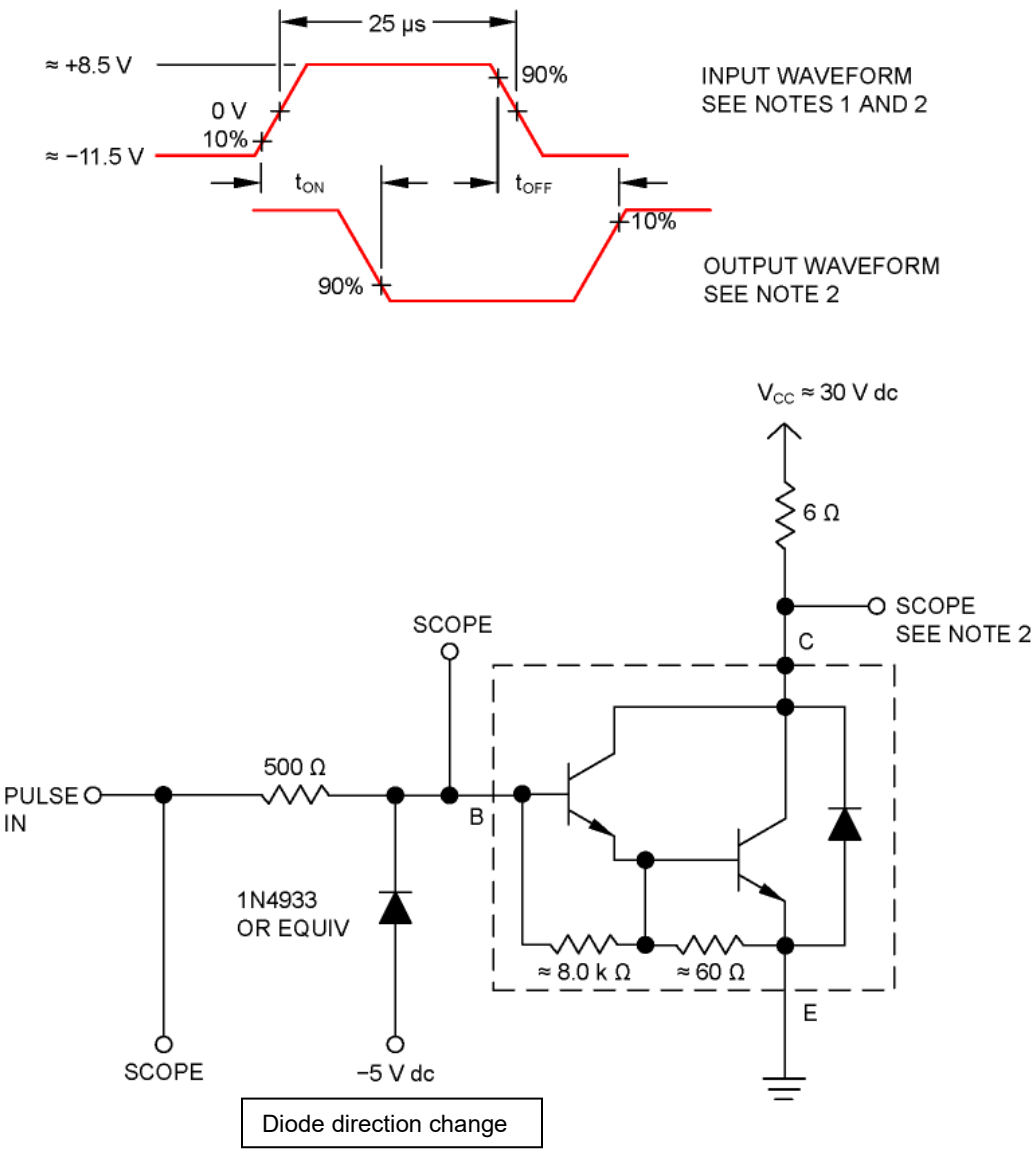
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* TABLE II. Group E inspection (all quality levels) - for qualification and requalification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices, c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles. | |
| Hermetic seal | 1071 | Fine and gross leak. | |
| * End-point electrical measurements | | See table I, subgroup 2 and 4.7 herein. | |
| Subgroup 2 | | | 45 devices, c = 0 |
| Intermittent operation life | 1037 | $V_{CB}$ ≥ 10 V dc, 6,000 cycles. | |
| * End-point electrical measurements | | See table I, subgroup 2 and 4.7 herein. | |
| Subgroup 4 | | | Sample size N/A |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 8 | | | 45 devices, c = 0 |
| Reverse voltage leakage stability | 1033 | Condition B. | |



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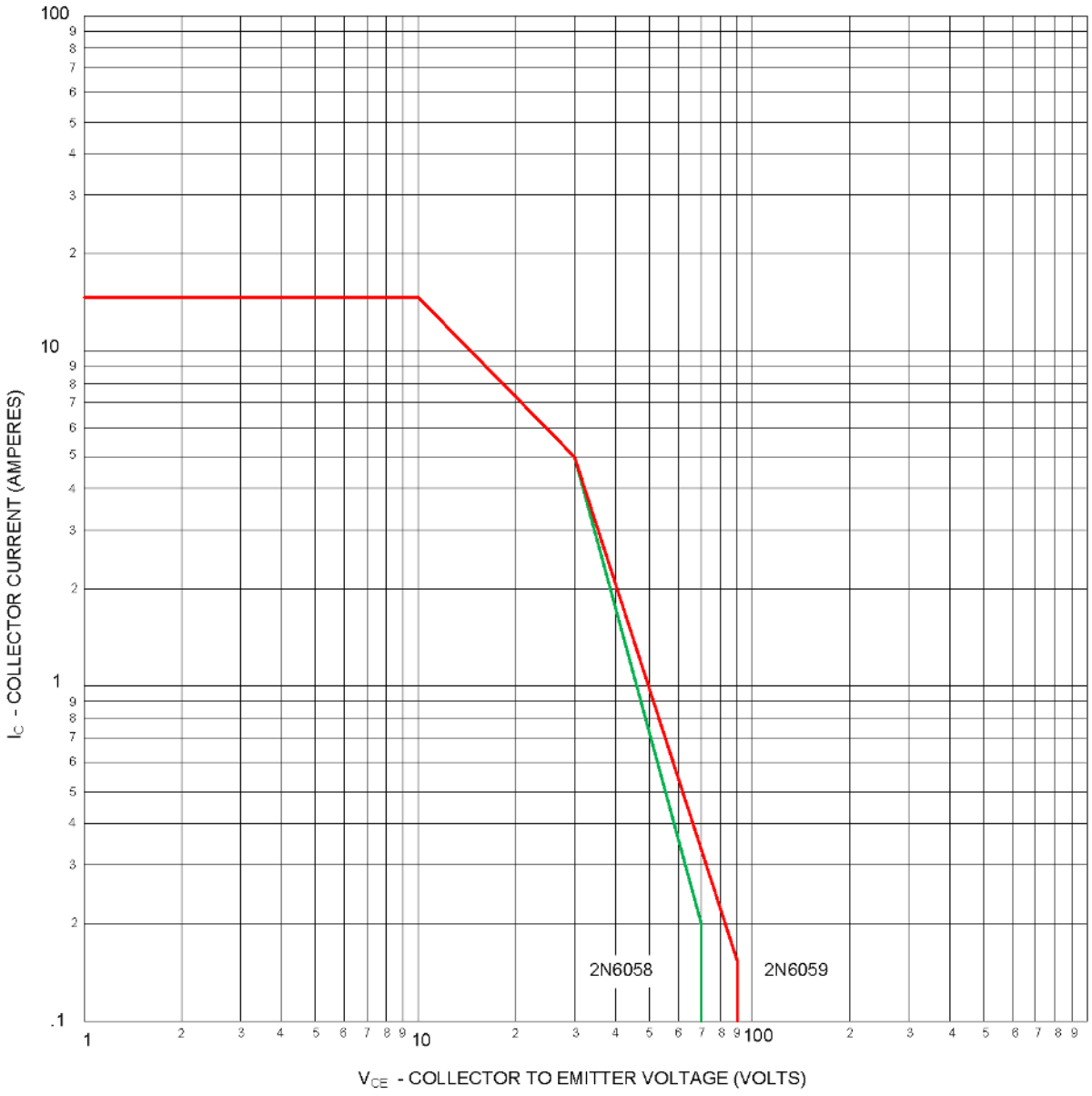

NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 20$ ns, $t_f \leq 20$ ns, $Z_{OUT} = 50$ ohms, PW = 25 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 2.0$ ns, $Z_{in} \geq 20$ kΩ, $C_{in} \leq 11.5$ pf.
3. Resistors shall be non-inductive types.
4. The dc power supplies may require additional by-passing in order to minimize ringing.

* FIGURE 3. Pulse response test circuit.



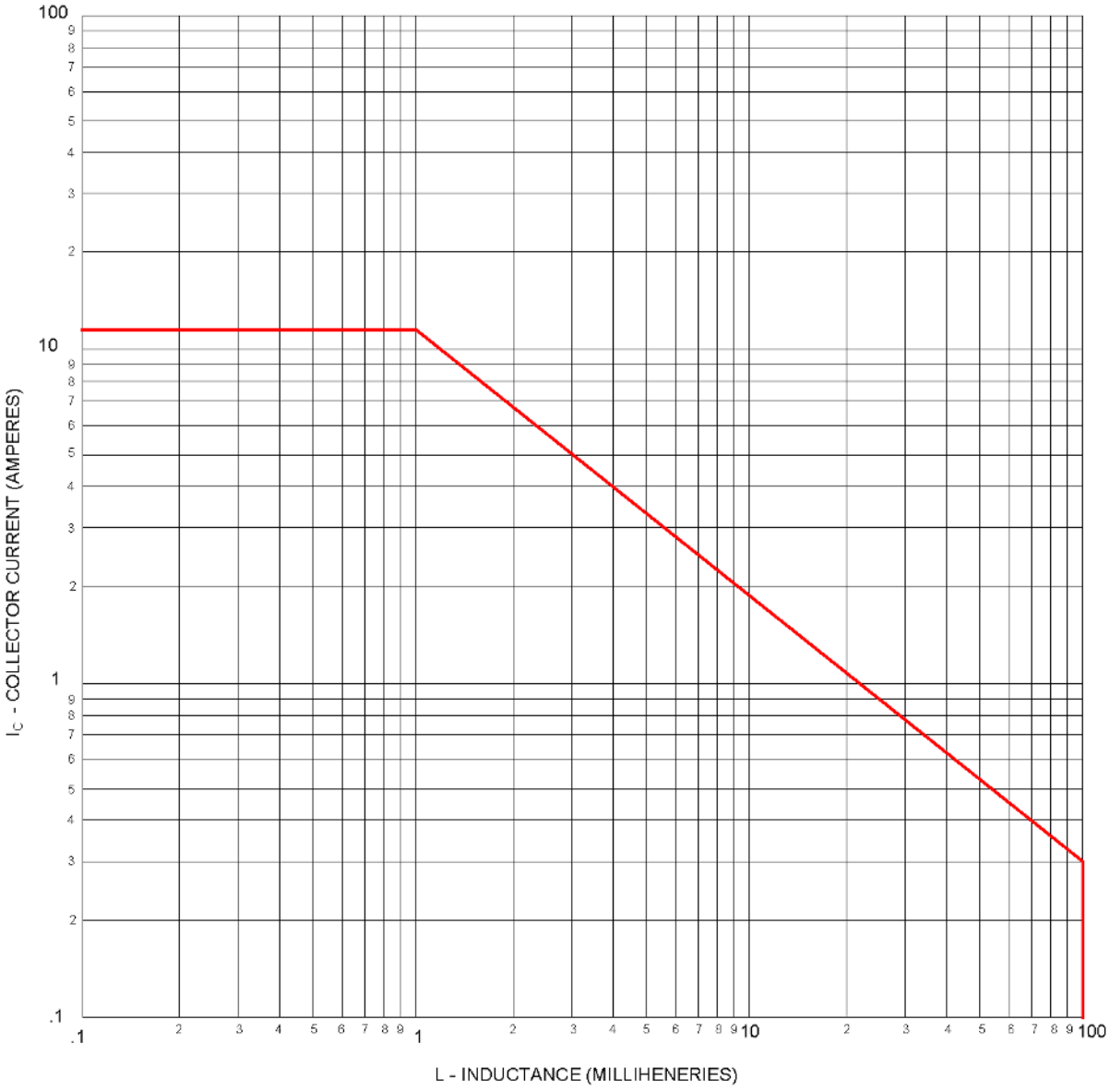
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* FIGURE 4. Maximum safe operating area graph (continuous dc).



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* FIGURE 5. Safe operating area for switching between saturation and cutoff (unclamped inductive load).



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.) The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

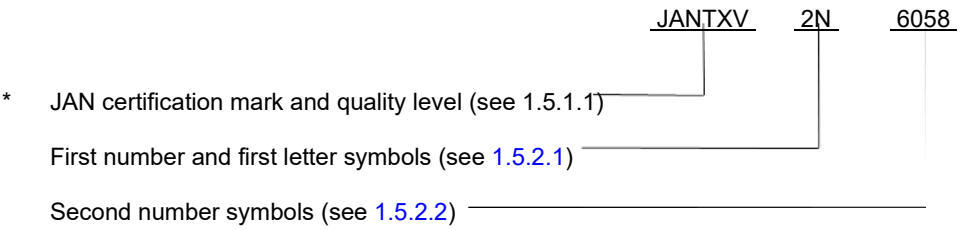
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

* 6.4 PIN construction example.

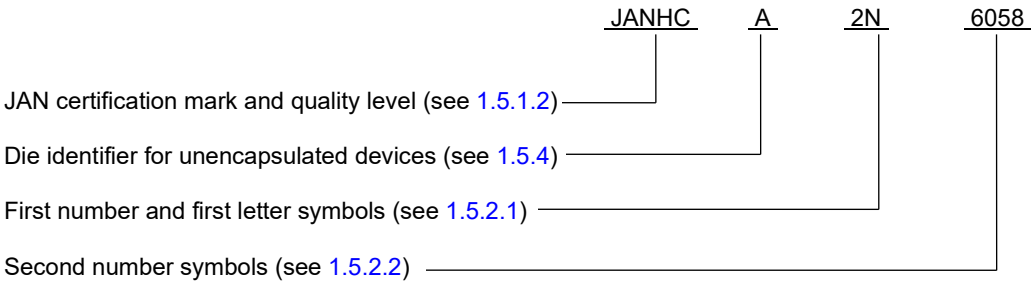
* 6.4.1 Encapsulated devices. The PINs for encapsulated devices are constructed using the following form.






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* 6.4.2 Unencapsulated devices. The PINs for unencapsulated devices are constructed using the following form.




* 6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

* 6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for devices of the base quality level | PINs for devices of the "TX" quality level | PINs for devices of the "TXV" quality level |
|---|---|---|
| JAN2N6058 | JANTX2N6058 | JANTXV2N6058 |
| JAN2N6059 | JANTX2N6059 | JANTXV2N6059 |

* 6.5.2 List of PINs for Unencapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for die quality level JANHC | PINs for die quality level JANKC |
|---|---|
| JANHCA2N6058 | JANKCA2N6058 |
| JANHCA2N6059 | JANKCA2N6059 |

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.



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6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Army − CR
Navy − SH
Air Force − 85
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Review activities:
Air Force − 19

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DLA − CC

(Project 5961−2024−083)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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